

LAB 5 TASKS

1. Retrieve all data from the employees table.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'homeuser' connection, which is expanded to show the 'HR' schema. The 'Tables (Filtered)' list includes COUNTRIES, DEPARTMENTS, EMPLOYEES, JOB_HISTORY, JOBS, LOCATIONS, and REGIONS. The 'Query Builder' pane shows the query 'select * from EMPLOYEES'. The 'Query Result' pane displays the fetched data.

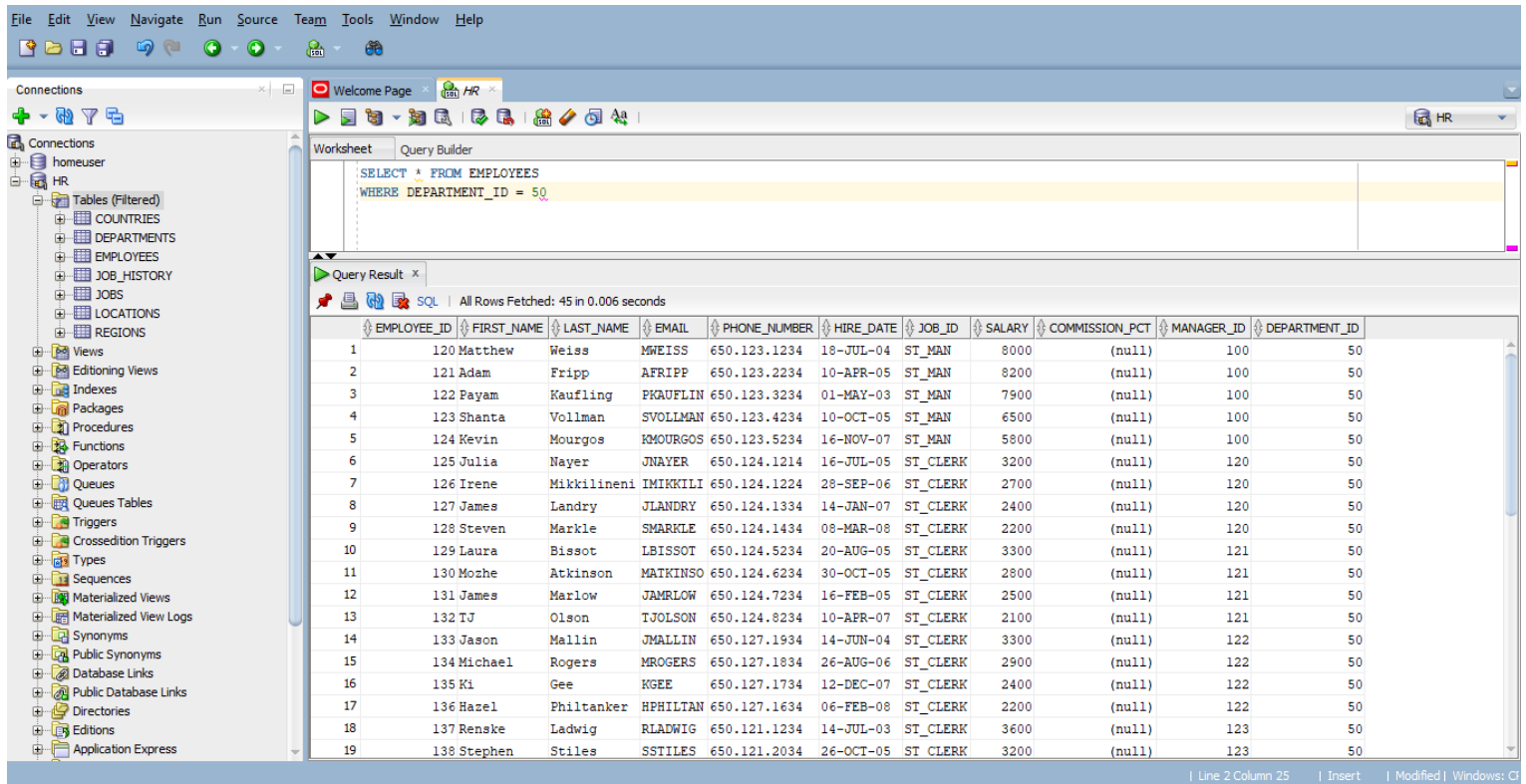
EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60
7	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60
8	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
9	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
10	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
11	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
12	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
13	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
14	113	Tjoe	Bane	TJORBANE	515.124.4567	07-DEC-07	FI_ACCOUNT	6800	(null)	108	100

2. Select first_name, last_name, and email of all employees.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'homeuser' connection, which is expanded to show the 'HR' schema. The 'Tables (Filtered)' list includes COUNTRIES, DEPARTMENTS, EMPLOYEES, JOB_HISTORY, JOBS, LOCATIONS, and REGIONS. The 'Query Builder' pane shows the query 'select first_name, last_name, email from EMPLOYEES'. The 'Query Result' pane displays the fetched data.

	FIRST_NAME	LAST_NAME	EMAIL
1	Ellen	Abel	EABEL
2	Sundar	Ande	SANDE
3	Mozhe	Atkinson	MATKINSO
4	David	Austin	DAUSTIN
5	Hermann	Baer	HBAER
6	Shelli	Baida	SBAIDA
7	Amit	Banda	ABANDA
8	Elizabeth	Bates	EBATES
9	Sarah	Bell	SBELL
10	David	Bernstein	DBERNSTE
11	Laura	Bissot	LBISSOT
12	Harrison	Bloom	HBLOOM
13	Alexis	Bull	ABULL
14	Anthony	Cabrio	ACABRIO

3. Retrieve details of employees who work in department 50.



Query Builder

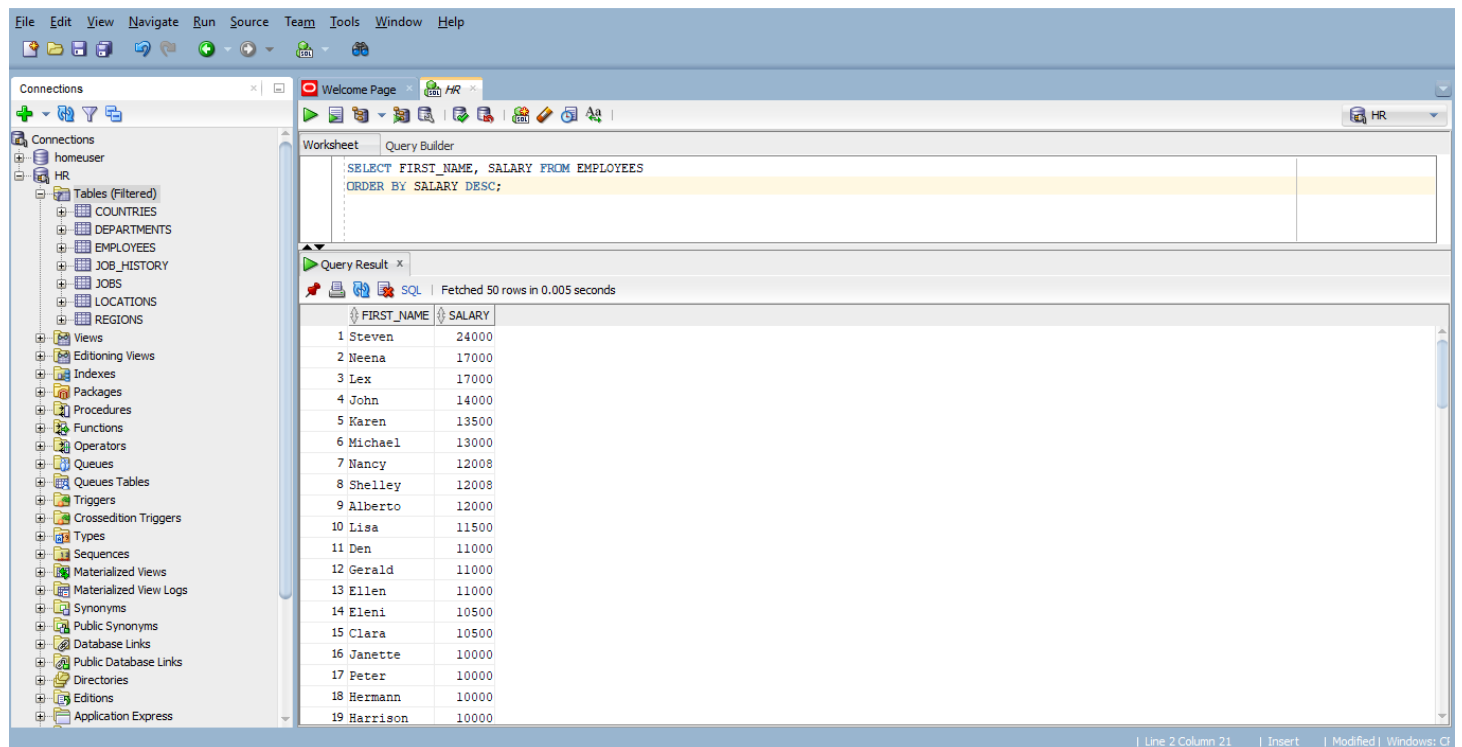
```
SELECT * FROM EMPLOYEES
WHERE DEPARTMENT_ID = 50
```

Query Result

All Rows Fetched: 45 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)	100	50
2	121	Adam	Fripp	AFRIPP	650.123.2234	10-APR-05	ST_MAN	8200	(null)	100	50
3	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
4	123	Shanta	Vollman	SVOLLMAN	650.123.4234	10-OCT-05	ST_MAN	6500	(null)	100	50
5	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
6	125	Julia	Nayer	JNAYER	650.124.1214	16-JUL-05	ST_CLERK	3200	(null)	120	50
7	126	Irene	Mikkilineni	IMIKKILI	650.124.1224	28-SEP-06	ST_CLERK	2700	(null)	120	50
8	127	James	Landry	JLANDRY	650.124.1334	14-JAN-07	ST_CLERK	2400	(null)	120	50
9	128	Steven	Markle	SMARKLE	650.124.1434	08-MAR-08	ST_CLERK	2200	(null)	120	50
10	129	Laura	Bissot	LBISSOT	650.124.5234	20-AUG-05	ST_CLERK	3300	(null)	121	50
11	130	Mozhe	Atkinson	MATKINSO	650.124.6234	30-OCT-05	ST_CLERK	2800	(null)	121	50
12	131	James	Marlow	JAMLOW	650.124.7234	16-FEB-05	ST_CLERK	2500	(null)	121	50
13	132	TJ	Olson	TJOLSON	650.124.8234	10-APR-07	ST_CLERK	2100	(null)	121	50
14	133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04	ST_CLERK	3300	(null)	122	50
15	134	Michael	Rogers	MROGERS	650.127.1834	26-AUG-06	ST_CLERK	2900	(null)	122	50
16	135	Ki	Gee	KGEE	650.127.1734	12-DEC-07	ST_CLERK	2400	(null)	122	50
17	136	Hazel	Philtanker	HPHILTAN	650.127.1634	06-FEB-08	ST_CLERK	2200	(null)	122	50
18	137	Renske	Ladwig	RLADWIG	650.121.1234	14-JUL-03	ST_CLERK	3600	(null)	123	50
19	138	Stephen	Stiles	SSTILES	650.121.2034	26-OCT-05	ST_CLERK	3200	(null)	123	50

4. Retrieve first_name and salary of employees, sorted by salary in descending order.



Query Builder

```
SELECT FIRST_NAME, SALARY FROM EMPLOYEES
ORDER BY SALARY DESC;
```

Query Result

Fetched 50 rows in 0.005 seconds

	FIRST_NAME	SALARY
1	Steven	24000
2	Neena	17000
3	Lex	17000
4	John	14000
5	Karen	13500
6	Michael	13000
7	Nancy	12008
8	Shelley	12008
9	Alberto	12000
10	Lisa	11500
11	Den	11000
12	Gerald	11000
13	Ellen	11000
14	Eleni	10500
15	Clara	10500
16	Janette	10000
17	Peter	10000
18	Hermann	10000
19	Harrison	10000

5. Retrieve the first 10 employees based on their hire date.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left shows the 'HR' schema. The 'Query Builder' pane shows the following SQL query:

```
SELECT * FROM
(SELECT * FROM employees ORDER BY hire_date)
WHERE ROWNUM <= 10;
```

The 'Query Result' pane shows the results of the query, displaying 10 rows of employee data:

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	
1	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
2	203	Susan	Mavris	SMAVRIS	515.123.7777	07-JUN-02	HR_REP	6500	(null)	101	40
3	206	William	Gietz	WGIEZT	515.123.8181	07-JUN-02	AC_ACCOUNT	8300	(null)	205	110
4	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(null)	101	110
5	204	Hermann	Baer	HBAER	515.123.8888	07-JUN-02	PR_REP	10000	(null)	101	70
6	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
7	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
8	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	FU_MAN	11000	(null)	100	30
9	122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	(null)	100	50
10	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	FU_CLERK	3100	(null)	114	30

6. Retrieve the names and salaries of employees whose salary is greater than 5000.

The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' pane shows the following SQL query:

```
SELECT FIRST_NAME, LAST_NAME, SALARY FROM EMPLOYEES
WHERE SALARY > 5000
```

The 'Query Result' pane shows the results of the query, displaying 19 rows of employee data:

FIRST_NAME	LAST_NAME	SALARY	
1	Steven	King	24000
2	Neena	Kochhar	17000
3	Lex	De Haan	17000
4	Alexander	Hunold	9000
5	Bruce	Ernst	6000
6	Nancy	Greenberg	12008
7	Daniel	Faviet	9000
8	John	Chen	8200
9	Ismael	Sciarra	7700
10	Jose Manuel	Urman	7800
11	Luis	Popp	6900
12	Den	Raphaely	11000
13	Matthew	Weiss	8000
14	Adam	Fripp	8200
15	Payam	Kaufling	7900
16	Shanta	Vollman	6500
17	Kevin	Mourgos	5800
18	John	Russell	14000
19	Karen	Partners	13500

7. Retrieve the names and salaries of employees whose salary is between 3000 and 7000.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Query Builder' window displays the following SQL query:

```
SELECT FIRST_NAME, LAST_NAME, SALARY FROM EMPLOYEES
WHERE SALARY BETWEEN 3000 AND 7000
```

The 'Query Result' pane shows 39 rows fetched in 0.006 seconds. The results are as follows:

	FIRST_NAME	LAST_NAME	SALARY
1	Bruce	Ernst	6000
2	David	Austin	4800
3	Valli	Pataballa	4800
4	Diana	Lorentz	4200
5	Luis	Popp	6900
6	Alexander	Khoo	3100
7	Shanta	Vollman	6500
8	Kevin	Mourgos	5800
9	Julia	Nayer	3200
10	Laura	Bissot	3300
11	Jason	Mallin	3300
12	Renske	Ladwig	3600
13	Stephen	Stiles	3200
14	Trenna	Rajs	3500
15	Curtis	Davies	3100
16	Oliver	Tuvault	7000
17	Sarath	Sewall	7000
18	David	Lee	6800
19	Sundar	Ande	6400

8. Retrieve details of employees who work in department 10, 20, or 30.

The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' window displays the following SQL query:

```
SELECT * FROM EMPLOYEES
WHERE DEPARTMENT_ID IN (10,20,30)
```

The 'Query Result' pane shows 9 rows fetched in 0.004 seconds. The results are as follows:

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	200	Jennifer	Whalen	JWHALEN	515.123.4444	17-SEP-03	AD_ASST	4400	(null)	101	10
2	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(null)	100	20
3	202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000	(null)	201	20
4	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(null)	100	30
5	115	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	PU_CLERK	3100	(null)	114	30
6	116	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	PU_CLERK	2900	(null)	114	30
7	117	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	PU_CLERK	2800	(null)	114	30
8	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(null)	114	30
9	119	Karen	Colmenares	KCOLMENA	515.127.4566	10-AUG-07	PU_CLERK	2500	(null)	114	30

9. Retrieve a list of unique department IDs from the employees table.

The screenshot shows the SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Query Builder' pane shows the query: `SELECT DISTINCT department_id FROM employees;`. The 'Query Result' pane displays the results of the query, showing 12 rows of distinct department IDs.

DEPARTMENT_ID
1
2
3
4
5
6
7
8
9
10
11
12

10. Retrieve the first_name, last_name, and department_name of all employees by joining the employees and departments tables

The screenshot shows the SQL Developer interface. The 'Query Builder' pane shows the query: `SELECT e.first_name, e.last_name, d.department_name FROM employees e JOIN departments d ON e.department_id = d.department_id;`. The 'Query Result' pane displays the results of the query, showing 19 rows of employee details joined with their respective department names.

FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1 Jennifer	Whalen	Administration
2 Pat	Fay	Marketing
3 Michael	Hartstein	Marketing
4 Sigal	Tobias	Purchasing
5 Karen	Colmenares	Purchasing
6 Shelli	Baida	Purchasing
7 Den	Raphaely	Purchasing
8 Alexander	Khoo	Purchasing
9 Guy	Himuro	Purchasing
10 Susan	Mavris	Human Resources
11 Kevin	Feeney	Shipping
12 Jean	Fleaur	Shipping
13 Adam	Fripp	Shipping
14 Timothy	Gates	Shipping
15 Ki	Gee	Shipping
16 Girard	Geoni	Shipping
17 Douglas	Grant	Shipping
18 Vance	Jones	Shipping
19 Payam	Kaufling	Shipping

11. Retrieve the first_name, last_name, and job_title of all employees by joining the employees and jobs tables.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Query Builder' pane displays the following SQL query:

```
SELECT e.first_name, e.last_name, j.job_title
FROM employees e
JOIN jobs j ON e.job_id = j.job_id;
```

The 'Query Result' pane shows the output of the query, which consists of 19 rows. The columns are FIRST_NAME, LAST_NAME, and JOB_TITLE.

FIRST_NAME	LAST_NAME	JOB_TITLE
1 William	Gietz	Public Accountant
2 Shelley	Higgins	Accounting Manager
3 Jennifer	Whalen	Administration Assistant
4 Steven	King	President
5 Neena	Kochhar	Administration Vice President
6 Lex	De Haan	Administration Vice President
7 Daniel	Faviet	Accountant
8 Luis	Popp	Accountant
9 John	Chen	Accountant
10 Ismael	Sciarra	Accountant
11 Jose Manuel	Urman	Accountant
12 Nancy	Greenberg	Finance Manager
13 Susan	Mavris	Human Resources Representative
14 Bruce	Ernst	Programmer
15 Alexander	Runold	Programmer
16 David	Austin	Programmer
17 Diana	Lorentz	Programmer
18 Vali	Pataballa	Programmer
19 Michael	Hartstein	Marketing Manager

12. Retrieve the first_name, last_name, department_name, and location_id of employees by joining employees, departments, and locations tables.

The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' pane displays the following SQL query:

```
SELECT e.first_name, e.last_name, d.department_name, l.location_id
FROM employees e
JOIN departments d ON e.department_id = d.department_id
JOIN locations l ON d.location_id = l.location_id;
```

The 'Query Result' pane shows the output of the query, which consists of 19 rows. The columns are FIRST_NAME, LAST_NAME, DEPARTMENT_NAME, and LOCATION_ID.

FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	LOCATION_ID
1 Ellen	Abel	Sales	2500
2 Sundar	Ande	Sales	2500
3 Mozhe	Atkinson	Shipping	1500
4 David	Austin	IT	1400
5 Hermann	Baer	Public Relations	2700
6 Shelli	Baida	Purchasing	1700
7 Amit	Banda	Sales	2500
8 Elizabeth	Bates	Sales	2500
9 Sarah	Bell	Shipping	1500
10 David	Bernstein	Sales	2500
11 Laura	Bissot	Shipping	1500
12 Harrison	Bloom	Sales	2500
13 Alexis	Bull	Shipping	1500
14 Anthony	Cabrio	Shipping	1500
15 Gerald	Cambrault	Sales	2500
16 Nanette	Cambrault	Sales	2500
17 John	Chen	Finance	1700
18 Kelly	Chung	Shipping	1500
19 Karen	Colmenares	Purchasing	1700

13. Retrieve all employees and their department details, including those who do not belong to any department (use an outer join).

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Worksheet' pane contains the following SQL query:

```
SELECT e.*, d.*
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

The 'Query Result' pane displays 50 rows of data. The columns are: EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER, HIRE_DATE, JOB_ID, SALARY, COMMISSION_PCT, MANAGER_ID, DEPARTMENT_ID, and DEPARTMENT_ID_1. The data shows employees and their corresponding department details.

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	DEPARTMENT_ID_1
1	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90	90
2	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90	90
3	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90	90
4	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	(null)	102	60
5	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60	60
6	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)	103	60	60
7	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)	103	60	60
8	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60	60
9	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100	100
10	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	(null)	108	100
11	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100	100
12	Ismail	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100	100
13	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100	100
14	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100	100
15	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	FU_MAN	11000	(null)	100	30	30
16	Alexander	Khoo	AKHOO	515.127.4562	18-MAY-03	FU_CLERK	3100	(null)	114	30	30
17	Shelli	Baida	SBAIDA	515.127.4563	24-DEC-05	FU_CLERK	2900	(null)	114	30	30
18	Sigal	Tobias	STOBIAS	515.127.4564	24-JUL-05	FU_CLERK	2800	(null)	114	30	30

14. Retrieve the first_name, last_name, and department_name of all employees, including those who are not assigned to any department, using a LEFT JOIN between the employees and departments tables (hint : use left join).

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Worksheet' pane contains the following SQL query:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
LEFT JOIN departments d ON e.department_id = d.department_id;
```

The 'Query Result' pane displays 50 rows of data. The columns are: FIRST_NAME, LAST_NAME, and DEPARTMENT_NAME. The data shows employees and their corresponding department names.

FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1 Jennifer	Whalen	Administration
2 Pat	Fay	Marketing
3 Michael	Hartstein	Marketing
4 Karen	Colmenares	Purchasing
5 Guy	Himuro	Purchasing
6 Sigal	Tobias	Purchasing
7 Shelli	Baida	Purchasing
8 Alexander	Khoo	Purchasing
9 Den	Raphaely	Purchasing
10 Susan	Mavris	Human Resources
11 Douglas	Grant	Shipping
12 Donald	OConnell	Shipping
13 Kevin	Feeney	Shipping
14 Alana	Walsh	Shipping
15 Vance	Jones	Shipping
16 Samuel	McCain	Shipping
17 Britney	Everett	Shipping
18 Sarah	Bell	Shipping
19 Randall	Perkins	Shipping

15. Retrieve the first_name, last_name, and department_name of all employees, including departments that have no employees, using a RIGHT JOIN between the employees and departments tables. (hint : use right join).

The screenshot displays the Oracle SQL Developer environment. On the left, the 'Connections' pane shows a tree view of database objects under the 'HR' schema, including tables like COUNTRIES, DEPARTMENTS, EMPLOYEES, and JOBS. The main workspace is divided into a 'Query Builder' tab and a 'Query Result' tab. The 'Query Builder' tab contains the following SQL query:

```
SELECT e.first_name, e.last_name, d.department_name
FROM employees e
RIGHT JOIN departments d ON e.department_id = d.department_id;
```

The 'Query Result' tab shows the output of the query, which consists of 50 rows. The first 19 rows are displayed in the table below:

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Pat	Fay	Marketing
3	Michael	Hartstein	Marketing
4	Sigal	Tobias	Purchasing
5	Karen	Colmenares	Purchasing
6	Shelli	Baida	Purchasing
7	Den	Raphaely	Purchasing
8	Alexander	Khoo	Purchasing
9	Guy	Himuro	Purchasing
10	Susan	Mavris	Human Resources
11	Kevin	Feeney	Shipping
12	Jean	Fleaur	Shipping
13	Adam	Fripp	Shipping
14	Timothy	Gates	Shipping
15	Ki	Gee	Shipping
16	Girard	Geoni	Shipping
17	Douglas	Grant	Shipping
18	Vance	Jones	Shipping
19	Payam	Kaufling	Shipping

The status bar at the bottom indicates 'Line 3 Column 63 | Insert | Modified | Windows: C'.